

Alyona Golz

Technical Project Engineer | Embedded Systems | UAV Technologies | C | Python | Linux

Location: Yerevan, Armenia

Professional Profile

Technical Project Engineer with a multidisciplinary background combining embedded systems, UAV technologies, software development, quality management, and business operations.

Experienced in development support, validation, and integration of autonomous systems, including UAV platforms, onboard computers, flight controllers, and industrial sensor solutions. Worked as part of the COEX engineering team on specialized unmanned aerial systems for scientific and industrial applications.

Strong technical foundation in C, C++, Python, Unix/Linux systems, algorithms, and low-level programming developed through the École 42 Computer Science program. Experienced in system debugging, hardware/software integration, technical documentation, and collaboration between engineering, product, and customer-facing teams.

Combines engineering mindset with business experience, enabling effective communication between technical teams, customers, and stakeholders.

Core Competencies

Embedded Systems & UAV Technologies

- UAV architecture and autonomous systems
- PX4 flight control ecosystem
- MAVLink communication protocol
- Flight controllers and onboard computers
- Raspberry Pi and Arduino platforms
- Hardware/software integration
- Sensor and payload integration
- System validation and troubleshooting

Software Engineering

- C and C++ programming
- Python development
- Unix/Linux environments

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- Algorithms and data structures
- Memory management
- Low-level programming
- Debugging and testing
- Git version control
- Makefile-based build systems

Systems & Infrastructure

- Linux administration
- Virtualization
- SSH and networking fundamentals
- Bash scripting
- Server configuration
- Database systems
- Cloud infrastructure fundamentals

Technical Operations

- Quality assurance
- Product validation
- Technical documentation
- Requirements analysis
- Cross-functional communication
- Process improvement

Professional Experience

[COEX \(Copter Express\)](#)

Technical Project & Quality Control Manager

2018 – 2022

COEX develops autonomous robotic and UAV solutions. Worked at the intersection of engineering, product development, quality assurance, and customer requirements.

Responsibilities:

- Supported engineering teams during development and validation of autonomous UAV platforms.
- Participated in hardware and software integration processes.
- Conducted functional testing, troubleshooting, diagnostics, and reliability verification.

- Analyzed technical issues and coordinated solutions between engineering teams and customers.
- Prepared technical documentation, testing reports, and operational materials.
- Improved quality control processes and supported product optimization

Selected Engineering Projects

Autonomous UAV Platform for Scientific Research (2018)

Supported development and validation of an autonomous UAV platform designed for Participated in development and hardware validation of a specialized UAV platform designed for Arctic environmental monitoring.

The system was created to deploy scientific buoys onto Arctic ice fields for long-term glacier melting research.

Technologies:

- PX4 flight control system
- COEX Pix Flight Controller
- Raspberry Pi onboard computer
- MAVLink communication protocol
- Embedded Linux
- Hardware integration and testing

Responsibilities:

- Supported integration of onboard electronics and payload systems.
- Performed system configuration and diagnostics.
- Participated in hardware validation and field preparation.
- Created technical documentation and test reports.

Result:

The platform successfully completed field testing on the Arctic shelf in September 2018.

Industrial Gas Monitoring UAV for Rosneft

UAV Platform with Industrial Sensor Payload (2020)

Participated in development and validation of an industrial UAV system designed for remote monitoring of oil and gas infrastructure.

The platform integrated an industrial gas analyzer with an autonomous aerial system.

Technologies:

- PX4 flight controller
- COEX Pix
- Raspberry Pi

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- MAVLink
- Embedded Linux
- Industrial sensor integration

Responsibilities:

- Supported integration of gas analysis equipment with UAV hardware.
- Performed system testing and validation.
- Assisted with troubleshooting communication between onboard modules.
- Verified reliable operation before deployment.

Result:

Successfully completed system integration and project validation for industrial deployment.

École 42 Software Engineering Projects

Born2beRoot

Linux Server Administration Project

Configured a virtual Linux server environment with security and service management.

Technologies:

- Rocky Linux
- VirtualBox
- Bash
- SSH
- LVM
- Lighttpd
- MariaDB
- PHP
- WordPress
- Fail2Ban

Skills developed:

- Linux administration
- Server configuration
- Security practices
- Automation scripting

minitalk

Unix Interprocess Communication

Developed a communication system between processes using Unix signals.

Technologies:

- C
- Unix signals
- Process management

Skills developed:

- IPC concepts
- Signal handling
- System-level programming

Previous Professional Experience

[Beeline Business Russia](#)

Key Account Manager — B2B Solutions (2012 – 2017)

Managed relationships with key corporate clients in Moscow and the Moscow region, providing ongoing support and consulting on telecommunications and technology solutions.

Responsibilities and achievements:

- Managed relationships with key corporate clients in Moscow and the Moscow region, providing ongoing support and consulting on telecommunications and technology solutions.
- Worked closely with corporate clients to understand business requirements and identify suitable technology solutions.
- Supported implementation and optimization of telecommunications services, including mobile communications, fixed-line services, and cloud solutions.
- Coordinated communication between customers, technical teams, and internal departments to resolve issues and improve service delivery.
- Developed strong expertise in enterprise customer operations, technical requirements analysis, and stakeholder communication.

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Usoft

Leading Specialist / Sales Lead Manager (2009-2011)

Worked with legal information systems and software solutions.

Responsibilities and achievements:

- Supported software adaptation for customer requirements.
- Analyzed client needs.
- Coordinated implementation processes.
- Combined technical understanding with customer support.

Education

École 42 Yerevan

Computer Science Program

2021 – 2024

Intensive project-based computer science education.

Main areas:

- C/C++ programming
- Algorithms
- Data structures
- Unix systems
- Operating systems
- Software architecture
- Team development methodology

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[Institute of Bioinformatics](#)

Python Programming Course

2020

Practical programming course focused on Python development, automation, and problem solving.

[Volga State University of Telecommunications and Informatics](#)

Master of Sociology

2005 – 2010

Developed analytical research methods, communication skills, and systems thinking.